

Preclinical In-Vitro Metabolic Clearance and Hepatic Cytochrome P450 Interactions of Cardioril

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ABSTRACT

The hepatic metabolic pathway of Cardioril was investigated in human liver microsomes (HLM) and Sprague-Dawley rat hepatocytes. Cardioril undergoes extensive oxidative metabolism predominantly mediated by CYP3A4 and CYP2C9 enzymes. In-vitro inhibition assays demonstrated an $IC_{50} > 50 \mu M$ for CYP2D6 and CYP1A2, indicating low potential for metabolic drug-drug interactions in animal models.

INTRODUCTION

Assessment of metabolic clearance and cytochrome P450 (CYP) inhibition kinetics is fundamental during early drug characterization. In the present investigation, we delineate the metabolic fate of Cardioril using cryopreserved human and rodent hepatocytes. This study contains no human clinical trials or individual patient case reports.

METHODS & EXPERIMENTAL PROCEDURES

In-vitro incubation of Cardioril (1–100 μM) was performed using pooled human liver microsomes (Corning Gentest) in the presence of NADPH-generating systems. Metabolite profiling was conducted via high-performance liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS). Rat hepatocyte viability was monitored via trypan blue exclusion.

RESULTS & DISCUSSION

Cardioril demonstrated rapid hepatic extraction with an intrinsic clearance (CL_{int}) of 42.5 $\mu L/min/mg$ protein in rat microsomes. No direct cytotoxic effects were observed in rodent hepatocytes up to 100 μM concentrations over 24-hour continuous exposure.

PHARMACOVIGILANCE ASSESSMENT

As this paper represents exclusively non-clinical in-vitro and animal laboratory research without identifiable human subjects, it does not meet regulatory criteria for ICSR submission under GVP Module VI Section VI.B.1. It is categorized as non-reportable literature.

REFERENCES

- Houston JB. Utility of in vitro drug metabolism data in predicting in vivo metabolic clearance. *Biochem Pharmacol.* 2019;47:1469-1479.